Recognition of Heartbeats Using Support Vector Machine Networks – A Comparative Study

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Abstract. The paper presents the comparison of performance of the individual and ensemble of SVM classifiers for the recognition of abnormal heartbeats on the basis of the registered ECG waveforms. The recognition system applies two different Support Vector Machine based classifiers and the ensemble systems composed of the individual classifiers combined together in different way to obtain the best possible performance on the ECG data. The results of numerical experiments using the data of MIT BIH Arrhythmia Database have confirmed the superior performance of the proposed solution

1 Introduction

The paper is concerned with the application of the Support vector Machine (SVM) classifiers for the recognition of heart rhythms on the basis of the registered ECG waveforms. The recognition system will be composed of the SVM used as the basic recognizing and classifying system and two different preprocessing stages resulting in different feature sets, composing the input signals to the classifier.

We will compare the performance of the heartbeat recognition system relying on the individual classifiers acting independently and on the ensemble of classifiers combined together to form the final recognizing system. The SVM networks have been used as the individual classifiers applied in the recognition of the ECG beats. The input vector **x** for these classifiers has been formed by the features following from the application of Hermite basis function expansion and of the higher order statistics (HOS) of the QRS complex of ECG. The individual classifiers will be combined together to form the ensemble network by applying the concept of voting. Three different methods of voting will be considered. The results of the numerical experiments for the recognition of 13 types of heart rhythms are presented and discussed.

2 The Individual SVM Classifiers for Heart Rhythm Recognition

To solve the problem of heartbeat recognition on the basis of the registered ECG waveform we have to generate the appropriate features describing the registered ECG waveform and apply them as the input signals to the classifier [1,2,3,4]. We have employed here the SVM classifier and two types of pre-processing of the data.

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2.1 Feature Extraction

The recognition of the patterns of the heart rhythms needs generation of the features, well characterizing patterns in a way enabling differentiation among different types of them. This is very important demand, since we observe great variation of signals among samples of the same type of beats, as is shown for example in the MIT BIH Arrhythmia Database [5].

In our work we have applied two methods of feature extraction: Hermite basis function expansion of the QRS complex and characterization of QRS by the cumulants of the second, third and fourth orders, well satisfying these general requirements [3]. In the numerical calculations, we presented the QRS segment of the ECG signal by 91 data points around the R peak (45 points before and 45 ones after).

In Hermite method we represent the QRS by the coefficients c_n of Hermite basis function expansion. If the analysed waveform is denoted by x(t) the expansion is defined by [2]

$$x(t) = \sum_{n=0}^{N-1} c_n \phi_n(t,\sigma) \tag{1}$$

where c_n are the expansion coefficients, σ - the width parameter and $\phi_n(t,\sigma)$ - the Hermite basis functions of nth order for n= 0, 1, 2, …, N-1. After some preliminary experiments we have applied 15 Hermite coefficients for ECG data representation.

In higher order statistics (HOS) approach [6] we represent the QRS complex by the values of the cumulants of the 2nd, 3rd and 4th orders [3], each calculated at five points distributed evenly within the QRS length (for the 3rd and 4th order cumulants the diagonal slices have been calculated). The application of HOS description reduces the variance of the registered ECG signals of each type and makes the recognition problem easier. For 91-element vector representation of the QRS complex the cumulants corresponding to the time lags of 15, 30, 45, 60 and 75 have been chosen.

Additionally we have added two temporal features: one corresponding to the instantaneous RR interval of the beat and the second representing the average RR interval of 10 preceding beats. In this way each beat, irrespective of its description, has been represented here by the 17-element feature vector, with the first 15 elements corresponding to either the higher order statistics or Hermite characterization of QRS complex, and the last two - the temporal features of the actual QRS signal.

2.2 Support Vector Machine Classifier

The Support Vector Machine (SVM) applied as the classifier is a linear machine [7] working in the high dimensional feature space formed by the linear or non-linear mapping of the n-dimensional input vector $\mathbf{x}$ into a *K*-dimensional feature space, usually of K>n through the use of a function $\varphi(\mathbf{x})$. The SVM network separates the data into two classes with maximum margin of separation. The hyperplane equation separating two different classes is given by $y(\mathbf{x}) = \mathbf{w}^T \boldsymbol{\varphi}(\mathbf{x}) = \sum_{j=1}^{K} w_j \varphi_j(\mathbf{x}) + w_0 = 0$, where $\boldsymbol{\varphi}(\mathbf{x}) = [\varphi_0(\mathbf{x}), \varphi_1(\mathbf{x}), ..., \varphi_K(\mathbf{x})]^T$ with $\varphi_0(\mathbf{x}) = 1$ and $\mathbf{w}$ – the weight vector of the network. The learning and testing modes of work are performed in SVM using so-called

kernel functions, satisfying the Mercer conditions [7,8]. The kernel $K(\mathbf{x},\mathbf{x}_i)$ is defined as the inner product of the vectors $\boldsymbol{\varphi}(\mathbf{x}_i)$ and $\boldsymbol{\varphi}(\mathbf{x})$, i.e., $K(\mathbf{x}_i,\mathbf{x}) = \boldsymbol{\varphi}^T(\mathbf{x}_i)\boldsymbol{\varphi}(\mathbf{x})$. The primal learning problem of SVM, formulated as the task of separating learning vectors $\mathbf{x}_i$ (i=1, 2, ..., p) into two classes of the destination values either $d_i=1$ (one class) or $d_i=-1$ (the opposite class), with the maximal separation margin is transformed to the so called dual problem of maximization of the function $Q(\boldsymbol{\alpha})$ with respect to the Lagrange multipliers α_i forming vector $\boldsymbol{\alpha}$ [7,8]. Dual problem solution results in the optimal values of the Lagrange multipliers, on the basis of which the output signal $y(\mathbf{x})$ of the SVM is determined as [7]

$$y(\mathbf{x}) = \sum_{i=1}^{N_s} \alpha_{si} d_i K(\mathbf{x}_{si}, \mathbf{x}) + w_o \tag{2}$$

If $y(\mathbf{x})>0$ the feature vector $\mathbf{x}$ belongs to the particular class and if $y(\mathbf{x})<0$ to the opposite one. The recognition of more classes is done in SVM by applying either "one against one" or "one against all" methods [8]. We have applied "one against one" approach, in which the SVM networks are trained to recognize between all combinations of two classes of data. For M classes we have to train M(M-1)/2 individual SVM networks. In the retrieval mode the vector $\mathbf{x}$ belongs to the class of the highest number of winnings in all combinations of classes.

3 The Integration Systems

Let us assume that there exist M channels of individual classifiers combined into one classifying system by the integrating part of the network. The measured signals of the ECG form the n-dimensional vector $\mathbf{x}_{in}$. This vector is transformed into different feature vectors $\mathbf{x}_i$ by the appropriate preprocessing blocks, forming the inputs to the classifiers. Each classifier has N binary outputs corresponding to N classes and the output signals of each classifier form the vector $\mathbf{y}_j$, for j=1, 2, …, M. The results of classifications of different classifiers may be combined together using different methods of integration. We will compare here 3 methods: the weighted voting (WV), Kulback-Leibler (K-L) and the modified Bayes approach (MB) [9].

The weighted voting combines the results $\mathbf{y}_i$ of M classifiers through the integrating matrix $\mathbf{W}$ to form one output vector $\mathbf{z}$ of the classifying system. The result of integration of all classifiers can be expressed by the relation

$$\mathbf{z}=\mathbf{W}\mathbf{y} \tag{3}$$

where $\mathbf{y} = \left[\mathbf{y}_1^T, \mathbf{y}_2^T, ..., \mathbf{y}_M^T\right]^T$. The position of the highest value element of $\mathbf{z}$ indicates the membership to the appropriate class. In adjusting the values of elements of the matrix $\mathbf{W}$ we have applied the minimization of the sum of squared error of the whole ensemble of the classifiers, measured on the learning data set [3]. This minimization leads to the solution expressed through the Moore-Penrose pseudoinverse in the following form $\mathbf{W} = \mathbf{D}\mathbf{Y}^+$, where $\mathbf{Y}$ is the $NM \times p$ matrix composed of p vectors $\mathbf{y}$ corresponding to p results of individual M classifications for learning data and $\mathbf{D}$ is

the appropriate $N \times p$ matrix formed by the destination vectors associated with each learning pair of data.

In Kullback-Leibler divergence [9] we calculate the ensemble probability μ_j supporting jth class given the actual input **x**, as the normalized arithmetic mean

$$\mu_j = \frac{1}{M} \sum_{i=1}^{M} d_{ij} \tag{4}$$

where d_{ij} means the probability of indicating jth class by ith classifier for the data of this class. This probability is determined in the testing mode for each classifier as the ratio of the number of victories of jth class to all possible indications in one against one mode of operation of SVM.

In modified naive Bayes combination [9] the ensemble probability μ_j for jth class is determined on the basis of results of testing the networks on learning data and is given in the form

$$\mu_j = \prod_{i=1}^{M} \frac{cm_{js_i}^{(i)} + 1/N}{n_j + 1} \tag{6}$$

where n_j is the number of elements in training set for class *j* and $cm_{js_i}^{(i)}$ is the element of the confusion matrix generated for learning data of ith classifier. The (j,s)th entry of the confusion matrix is the number of elements of the data set whose true class label was *j* and were assigned by ith classifier to sth class.

4 The Results of Numerical Experiments

The experiments have been performed for 13 types of heartbeats contained in MIT BIH Arrhythmia Database [5]. In this data base there are ECG waveforms of 12 types of abnormal beats: left bundle branch block (L), right bundle branch block (R), atrial premature beat (A), aberrated atrial premature beat (a), nodal (junctional) premature beat (J), ventricular premature beat (V), fusion of ventricular and normal beat (F), ventricular flutter wave (I), nodal (junctional) escape beat (j), ventricular escape beat (E), supraventricular premature beat (S) and fusion of paced and normal beat (f) and the waveforms corresponding to the normal sinus rhythm (N). We have used 12785 heart rhythms (6690 for learning and 6095 for testing). The SVM networks of radial Gaussian kernels have been applied and one against one strategy. The optimal values of regularization constant C and parameter σ of Gaussian function have been adjusted using cross validation approach for the learning data (C=100, σ=0.5). They have been set constant for all SVM networks.

Table 1 presents the comparison of total results achieved by individual SVM based classifiers (Hermite and HOS preprocesing), as well as by the ensembles of both classifiers for the learning and testing data (not used in learning). The rows labelled as WV, K-L and MB denote the results of integration of classifier results. The misclassification rate has been calculated as the mean of errors of each class recognition. There is a visible improvement of the results of testing after application of ensemble of

Table 1. The comparison of the average misclassification rate of testing different SVM solutions for the recognition of 13 types of the heartbeats

Classification method	Total number of learning errors	Average relative learning error	Total number of testing errors	Average relative testing error
Hermite	103	3.02%	172	5.23%
HOS	173	5.19%	216	6.28%
WV	74	1.83%	159	4.09%
K-L	108	2.27%	147	3.77%
MB	73	1.80%	171	4.39%

Table 2. The confusion matrix of the ensemble classifier system for 13 types of rhythms of the testing data

Heart type	A	a	f	E	F	j	I	J	L	N	R	S	V	Total
A	399	3	1	0	1	5	1	0	1	16	0	0	6	433
a	2	57	0	0	1	0	0	0	0	2	0	0	3	65
f	0	0	195	0	0	0	0	1	6	1	0	0	0	203
E	0	0	0	48	0	0	0	0	0	0	0	0	0	48
F	0	0	0	0	358	0	0	0	0	2	1	0	5	366
j	2	0	1	1	1	95	0	0	0	5	0	0	0	105
I	0	0	0	0	0	0	193	0	1	1	0	1	2	198
J	0	0	0	0	1	0	0	36	0	3	0	0	0	40
L	1	0	1	0	0	0	0	0	488	5	0	0	5	500
N	10	3	1	0	1	4	0	1	3	1959	2	0	1	1985
R	1	0	0	1	0	0	0	1	0	1	395	0	1	400
S	1	0	0	0	0	1	6	0	0	0	2	511	0	521
V	2	1	1	0	7	0	0	0	1	5	0	0	1214	1231
Total	418	64	200	50	370	105	200	39	500	2000	400	512	1237	6095

classifiers. For the best Kulback-Leibler (K-L) integration it is more than 20% in relation to the best individual classifier. Table 2 presents the confusion matrix of classification for the best ensemble system.

The diagonal entries of this matrix represent right recognition of the beat type and the off diagonal – the misclassifications. Each column presents how the beats of particular type have been classified. The row indicates which beats have been classified as the type mentioned in this row.

In practice the most dangerous case is when the ill person is diagnosed as the healthy one (false negative diagnosed patient). To deal with such case we have introduced the parameter DUV (dangerous uncertainty value), defined as the ratio of the number of all false negative diagnosed patients to the total number of misclassifications. Table 3 presents the values of this parameter for the individual SVM classifiers and for the integrated system. There is an evident improvement of the quality of classification, both in learning and in the testing mode.

Table 3. The comparison of DUV values for different SVM based classifiers

Quality measure	Learning data			Testing data		
	HOS	HER	Integration	HOS	HER	Integration
DUV	31.79%	38.83%	27.03%	18.52%	23.84%	17.68%

5 Conclusions

The paper has presented the application of the single SVM based classifier and the ensemble of classifiers for the recognition of heartbeats on the basis of ECG waveforms. The numerical results performed on the MIT BIH AD examples have shown that the SVM networks combined into the ensemble system composed of individual classifiers brings in significant improvement of the classification results, especially reduction of the most dangerous misclassification cases. The experimental results have shown that instead of designing one high performance classifier we can build a number of classifiers, each of possibly not superb performance and as a results we get the classifying system of significantly higher quality.

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